

**MID-SEMESTER COMPUTER NETWORKING
EXAMINATION REPORT**

5-Router Topology with Metric Calculations



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CHAPTER I

INTRODUCTION

1.1. Background

In computer networking, routers play a crucial role in connecting multiple networks and ensuring that data is transmitted efficiently between them. As the complexity of network infrastructure increases, understanding how routers communicate and determine the best path becomes essential for network engineers. One of the key aspects of routing is the calculation of metrics, which are numerical values used to evaluate the efficiency or cost of a particular route.

Through this practicum, students will design and analyze a five-router topology using Cisco Packet Tracer. The activity focuses on understanding how routing metrics affect the selection of paths in data transmission and how network performance can be optimized through proper configuration. This practicum also helps students develop technical skills in IP addressing, network simulation, and troubleshooting.

1.2. Objective

The main objective of this practicum is to build and configure a five-router topology and perform metric calculations to determine the best routing paths. Specifically, the practicum aims to:

1. Understand the concept of routing metrics and their impact on network performance.
2. Configure static or dynamic routes among five interconnected routers.
3. Calculate and compare routing metrics to determine the most efficient path.

CHAPTER II

THEORETICAL BASIS

2.1. Network Topology Overview

A mesh is a network configuration in which every device (or router) is directly connected to several, or in some cases all, other devices in the network. This arrangement provides high redundancy and reliability, because data can travel through multiple alternative paths if one link fails. Mesh networks are commonly used in backbone infrastructures, large-scale enterprise systems, and laboratory simulations that require continuous connectivity.

In this practicum, a five-router mesh topology is implemented using *Cisco Packet Tracer*. Each router is interconnected to create multiple communication routes between subnets. This design allows students to observe how routers select the best possible path based on metric values, and how the network continues functioning even if one link becomes unavailable.

2.2. Routing Concepts and Metrics

Routing is the process of determining the most efficient path for data packets to travel from a source network to a destination network. Each router maintains a routing table containing destination addresses, next-hop information, and corresponding metric values. Routing methods are divided into two main categories:

1. Static Routing

All routes are manually configured by the network administrator. It offers full control but requires manual updates if the topology changes.

2. Dynamic Routing

Routers automatically exchange routing information through protocols such as RIP, OSPF, or EIGRP, which use metrics to select optimal paths.

A metric represents the “cost” or efficiency of a particular route. Common metrics include:

1. Hop Count
Number of routers a packet passes through.
2. Bandwidth
Data-carrying capacity of a link; higher bandwidth equals lower cost.
3. Delay
Time required for data to traverse the link.
4. Load
Current traffic level and link stability.
5. Administrative Cost
Manually assigned preference value.

In a mesh topology, multiple redundant paths exist between routers, so metric calculation is vital to ensure that routers choose the shortest or least-cost route.

2.3. Routing Protocols Used in Mesh

Routing protocols determine how routers communicate to share information about network topology and metrics. They can be categorized into two types:

1. Distance Vector Protocols

Such as RIP, which use hop count as the main metric and periodically broadcast their entire routing tables to neighbors. They are simpler but less efficient for large or dense mesh networks.

2. Link-State Protocols

Such as OSPF and IS-IS, which rely on a link-state database and cost-based metrics derived from bandwidth. Each router independently computes the shortest-path tree using algorithms like Dijkstra's Algorithm, making them more suitable for complex mesh topologies.

In this practicum, the metric calculations are analyzed to show how routers in a mesh topology determine the most efficient route between nodes and how different metrics influence the final routing decision.

CHAPTER III

METHODOLOGY

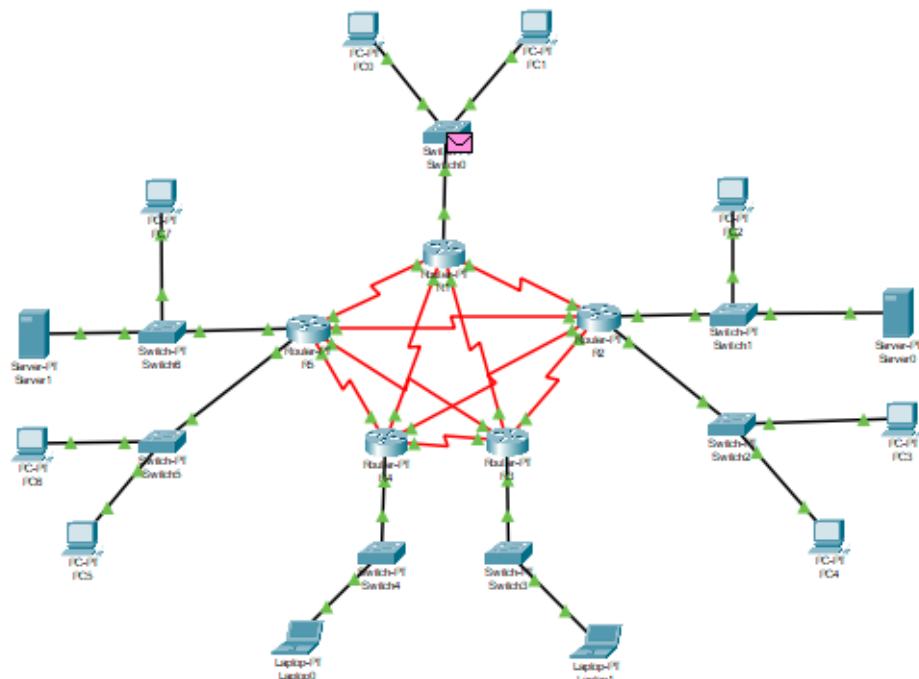
3.1. Tools and Materials

The practicum was conducted using both hardware and software components that support network simulation and configuration. The tools and materials used include:

1. Software Tools
Cisco Packet Tracer (network simulation software)
2. Network Devices (Simulated)
 - 5 Routers (PT-Router)
 - 5 Switches (optional, for end devices)
 - PCs or end devices for testing connectivity
 - Ethernet cables (Serial DCE/DTE and Copper Straight-Through)

3.2. Network Design and IP Addressing Scheme

The designed network uses a mesh topology connecting five routers labeled R1, R2, R3, R4, and R5. In a mesh configuration, each router is interconnected with several others, ensuring multiple redundant paths for data communication.



Link	Network (/30)	Side A (Router/IF)	IP	Side B (Router/IF)	IP
R1–R2	10.10.10.0/30	R1 S2/0	10.10.10.1	R2 S3/0	10.10.10.2
R2–R3	11.11.11.0/30	R2 S2/0	11.11.11.1	R3 S3/0	11.11.11.2
R3–R4	12.12.12.0/30	R3 S2/0	12.12.12.1	R4 S3/0	12.12.12.2
R4–R5	13.13.13.0/30	R4 S2/0	13.13.13.1	R5 S3/0	13.13.13.2
R1–R5	14.14.14.0/30	R5 S2/0	14.14.14.1	R1 S3/0	14.14.14.2
R1–R3	15.15.15.0/30	R1 S4/0	15.15.15.1	R3 S5/0	15.15.15.2
R1–R4	16.16.16.0/30	R1 S5/0	16.16.16.1	R4 S5/0	16.16.16.2
R2–R5	17.17.17.0/30	R5 S4/0	17.17.17.1	R2 S5/0	17.17.17.2
R2–R4	18.18.18.0/30	R2 S4/0	18.18.18.1	R4 S4/0	18.18.18.2
R3–R5	19.19.19.0/30	R5 S5/0	19.19.19.1	R3 S4/0	19.19.19.2

3.3. Configuration Steps

The configuration procedure follows systematic steps to ensure correct routing setup and metric calculation:

1. Place five routers (R1–R5) in the workspace and connect them using serial based on the mesh design.
2. Configure IP addresses for each router interface according to the IP addressing scheme.
3. Use dynamic routing protocol (RIP) to enable communication between routers.
4. Test connections between routers using the ping command to ensure all routes work properly.
5. Observe routing tables and calculate metrics (hop count for RIP) to determine the most efficient path between routers.

CHAPTER IV

RESULTS AND DISCUSSION

4.1. Simulation Result

The practical setup was successfully implemented in Cisco Packet Tracer using a 5-router mesh topology (R1–R5). Each router was connected through serial interfaces, forming multiple redundant communication paths. Each router was configured with unique IP addresses according to the subnet plan from Chapter III. Static or dynamic routing (such as RIP v2 or OSPF) was applied to enable communication between all subnets.

After configuration, connectivity was tested using the `ping` command from various PCs and servers connected to different routers. The test results showed that all devices were able to communicate successfully, confirming that the routing tables were properly built and that the mesh network provided full redundancy.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit ^
	Successful	PC7	PC2	ICMP		0.000	N	0	(ec
	Successful	PC7	Laptop0	ICMP		0.010	N	1	(ec
	Successful	PC7	PC4	ICMP		0.493	N	2	(ec

4.2. Metric Calculation Result

Source Router	Destination Network	Next Hop (Router/IP)	RIP Metric (Hops)
R1	192.168.2.0/24	R2 – 10.10.10.2	1
	192.168.3.0/24	R2 – 10.10.10.2	1
	192.168.4.0/24	R3 – 15.15.15.2	1
	192.168.5.0/24	R4 – 16.16.16.2	1
	192.168.6.0/24	R5 – 14.14.14.1	1
	192.168.7.0/24	R5 – 14.14.14.1	1
R2	192.168.1.0/24	R1 – 10.10.10.1	1
	192.168.4.0/24	R3 – 11.11.11.2	1
	192.168.5.0/24	R4 – 18.18.18.1	1
	192.168.6.0/24	R5 – 17.17.17.1	1

	192.168.7.0/24	R5 – 17.17.17.1	1
R3	192.168.1.0/24	R1 – 15.15.15.1	1
	192.168.2.0/24	R2 – 11.11.11.1	1
	192.168.3.0/24	R2 – 11.11.11.1	1
	192.168.5.0/24	R4 – 12.12.12.2	1
	192.168.6.0/24	R5 – 19.19.19.2	1
	192.168.7.0/24	R5 – 19.19.19.2	1
R4	192.168.1.0/24	R1 – 16.16.16.1	1
	192.168.2.0/24	R2 – 18.18.18.2	1
	192.168.3.0/24	R2 – 18.18.18.2	1
	192.168.4.0/24	R3 – 12.12.12.1	1
	192.168.6.0/24	R5 – 13.13.13.2	1
	192.168.7.0/24	R5 – 13.13.13.2	1
R5	192.168.1.0/24	R1 – 14.14.14.2	1
	192.168.2.0/24	R2 – 17.17.17.2	1
	192.168.3.0/24	R2 – 17.17.17.2	1
	192.168.4.0/24	R3 – 19.19.19.1	1
	192.168.5.0/24	R4 – 13.13.13.1	1

Based on the routing tables generated during the simulation, every router was able to reach all destination networks within one hop. This occurs because, in a full mesh topology, each router is directly connected to all other routers. Therefore, there are no intermediate routers between any two nodes. For example, R1 reaches the networks 192.168.2.0/24 and 192.168.3.0/24 through R2 (10.10.10.2), 192.168.4.0/24 through R3 (15.15.15.2), 192.168.5.0/24 through R4 (16.16.16.2), and both 192.168.6.0/24 and 192.168.7.0/24 through R5 (14.14.14.1), each with a metric value of 1.

Similarly, all other routers (R2–R5) have routes to every network with a RIP metric of 1 hop, confirming that each router is directly linked. Mathematically, the RIP metric for each route is calculated as:

$$\text{RIP Metric} = \text{Number of Hops}$$

Because each router in this mesh topology has a direct point-to-point connection, the number of hops is one for every route. For example:

Metric (R1-R2) = 1 ; Metric (R1-R3) = 1

If a router were connected through another router (for example, R1 → R2 → R3), the hop count would increase by one for each additional router along the path. Thus, the total metric would be 2 hops.

From the results, the RIP metric calculation demonstrates that a mesh topology provides optimal connectivity, with every router having equal and minimal hop counts to all destinations. This ensures high availability, low latency, and efficient routing throughout the network.

CHAPTER V

CLOSING

5.1. Conclusion

The practicum on 5-Router Mesh Topology with Metric Calculations was successfully carried out using *Cisco Packet Tracer*. The network design demonstrated how multiple routers can be interconnected to create a full mesh topology that provides complete redundancy and high reliability. Each router was configured with a unique IP address, and routing information was dynamically exchanged using the Routing Information Protocol (RIP).

The metric calculation results showed that every route in the mesh topology had a hop count value of 1, indicating that all routers are directly connected to one another. This configuration resulted in optimal routing performance, low latency, and efficient data transmission. The practicum also illustrated that RIP uses a simple metric based solely on hop count.

Overall, this practicum provided valuable understanding of the principles of dynamic routing, metric calculation, and network topology design. Students were able to comprehend the relationship between topology structure and routing performance through direct configuration, simulation, and analysis.